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Sovereign vs. Closed: Hanzo AI versus Centralized Cloud AI for Defense and Regulated Workloads

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Abstract

We compare two structurally different ways to obtain frontier AI: a closed, centrally hosted model accessed as a metered cloud service (typified by Anthropic's Claude), and a sovereign, self-hostable stack with owned model weights (Hanzo). The distinction is not primarily one of model quality—closed frontier models are excellent—but of *deployment model and control*. For defense and regulated workloads, the binding constraints are data residency, operation in disconnected and contested environments, supply-chain provenance, post-quantum security, auditability, and cost at scale. A closed cloud service cannot satisfy several of these by construction: it cannot run air-gapped, its weights cannot be inspected or owned, and sensitive data must leave the enclave to be processed. We argue that for these workloads the relevant choice is between foreign open-weight models and an American open-source alternative, and we position Hanzo as the latter—owning both the model (the Zen family) and the cryptography (a production post-quantum stack). We give a dimension-by-dimension comparison and a decision framework for when each model of consumption fits.

Executive Summary. This is not a claim that our model is smarter than Claude. It is a claim about *where* and *how* you can run it. Claude is an excellent frontier model delivered as a managed cloud service: you call it over the internet, your data goes to the vendor, you cannot see or own the weights, and it cannot run on a disconnected device. That is the right product for many commercial uses and the wrong one for a workload that cannot leave a controlled environment. Hanzo is the opposite shape: open, owned-weight models you run on your own hardware—in a data center, on a vehicle, or air-gapped—with post-quantum security built in, your data never leaving, and cost driven down because you are not paying a per-token markup or a hyperscaler margin. America's frontier AI has narrowed to two closed labs; the only openly available frontier weights today are largely foreign. For defense and regulated buyers who need open, sovereign, controllable AI, that makes Hanzo the American alternative to both the closed duopoly and foreign open weights.

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1 The Real Axis of Comparison

Discussions of "which AI is better" usually compare benchmark scores. For commercial chat and coding that is a reasonable proxy. For defense and regulated deployment it is the wrong axis. A model that scores two points higher on a reasoning benchmark is of no use inside a controlled environment if it can only be reached by sending the data outside that environment.

The decisive axis is the *deployment and trust model*: who holds the weights, where inference physically runs, whether the data leaves the enclave, whether the system works without connectivity, what cryptography protects it, and what it costs at scale. On this axis a closed cloud service and a sovereign self-hosted stack are not competitors with different scores—they are different categories of thing. Claude is the leading example of the first category; Hanzo is built as the second.

We state plainly what closed cloud AI does well: it is operationally simple (no infrastructure to run), it tracks the frontier of raw capability, and its provider invests heavily in alignment and safety research. None of that is in dispute. The argument here is narrow and structural: several requirements that are mandatory for defense and regulated workloads cannot be met by a model you do not hold and cannot run yourself.

2 Dimension-by-Dimension

Dimension	**Closed cloud AI (e.g. Claude)**	**Hanzo (sovereign OSS)**
Model weights	Not disclosed; not transferable	Owned (Zen family, 0.8B–1T+); inspectable
Where inference runs	Vendor cloud only	Your hardware: data center, edge, air-gapped
Your data	Leaves the enclave to the vendor	Never leaves; stays where it lives
Disconnected / DDIL	Not possible (requires connectivity)	First-class (offline on embedded ARM)
Provenance	Vendor-controlled	American-owned, full provenance
Post-quantum security	Not a product property	Native (ML-KEM-768 + ML-DSA-65)
Auditability	Black box	Open stack; cryptographically signed actions
Customization	Prompting / limited fine-tune via API	Full fine-tune on your data, in your enclave
Cost model	Per-token metered	Owned + self-hosted; no markup or cloud margin
Lock-in	High (API + data gravity)	Low (you hold the stack)

Table 1: Closed cloud AI versus a sovereign self-hostable stack. The comparison is of deployment model, not of raw benchmark quality.

2.1 Data residency and the enclave boundary

The single most consequential difference is the enclave boundary. To use a closed cloud model, the data being reasoned over must cross the boundary to the vendor. For CUI, engineering files, procurement records, or operational data this is often simply prohibited. Hanzo runs the model inside the boundary, so the data never crosses it. This is not a policy preference that can be waived with a contract clause; it is a property of where computation physically happens.

2.2 Disconnected and contested operation

A cloud service is unavailable the moment the link drops or is jammed. At the tactical edge, on a ship, or in an air-gapped facility there is no link to call. Hanzo's models are quantized to run fully offline on hardware ranging from a laptop to an embedded processor, so capability persists in exactly the environments where a cloud service cannot operate at all.

2.3 Provenance and the supply chain

For an open, controllable alternative to a closed cloud model, today's practical options are foreign open-weight models or Hanzo. Defense buyers care about who produced the weights and whether they can be trusted and inspected. Hanzo's position is specific: an American-owned, open-source frontier stack with owned weights, so an organization can have open, auditable AI without depending on either the closed domestic duopoly or foreign-origin weights.

2.4 Post-quantum security

Security against a future quantum adversary is a property of the whole stack, not a feature of a chat API. Hanzo's transport, identity, and key management are post-quantum by construction (all three NIST standards in production, formally verified). A closed chat service does not expose this as something the customer controls.

2.5 Cost at scale

A metered API charges per token on every inference, forever, with the vendor's margin baked in. Owning the model and self-hosting removes both the per-token markup and the hyperscaler margin; combined with an efficient transport layer (the ZAP protocol's 91% memory and 96% energy reductions) this is the mechanism behind roughly an order-of-magnitude lower infrastructure cost at scale. A companion case study documents a regulated client whose cloud spend fell approximately 97% after migrating to this stack.

2.6 Agentic throughput: one engineer, a team's output

A decisive data point sits behind this whole comparison. Hanzo's founding CEO/CTO, Zach Kelling, was—by public usage tracking (`claudecount.com`)—the **#1 user of Anthropic by token count worldwide in 2025**. That throughput was not spent on demos: on the flagship regulated production build, a single founding engineer, multiplied by the agentic stack, authored the substantial majority of the production code—thousands of commits in a matter of weeks—output that conventionally requires a sizable team. The point for this paper is what came next. That same throughput now runs on Hanzo's *owned*, self-hostable stack rather than a metered cloud API. The capability that topped a closed provider's global usage is exactly the capability Hanzo packages as a sovereign, owned alternative—so a buyer gets the productivity without the per-token meter, the data egress, or the dependency. That is the comparison in one line: same frontier-grade agentic output, but owned instead of rented.

3 When Each Model Fits

This is a decision framework, not a verdict. Closed cloud AI is the right choice when the workload is non-sensitive, connectivity is reliable, operational simplicity outweighs control, and there is no requirement to own or inspect the model. A sovereign self-hosted stack is the right—often the only—choice when any of the following hold: the data cannot leave a controlled environment; the system must work disconnected or in contested conditions; provenance and auditability are required; post-quantum protection is mandated; or per-token cost at scale is untenable. Defense and regulated workloads concentrate in the second column. That is the gap Hanzo is built to fill.

4 Conclusion

The honest comparison is not "Hanzo's model beats Claude." It is that a closed cloud model and a sovereign self-hostable stack are different categories, and the requirements of defense and regulated workloads—data residency, disconnected operation, provenance, post-quantum security, auditability, and cost—select the second category by construction. Within that category the choice narrows to foreign open weights or an American open-source alternative. Hanzo is built to be that alternative: own the model, own the cryptography, run it where the data lives.